

Particulate Matter

Monitoring & Health Effects

Issue 2026-08-09 | Entry window 9 August 2026 | 12 records in scope

12RECORDS IN SCOPE
(16 REJECTED)**8**SENSING / EXPOSURE
RECORDS (67%)**6**SUBTOPIC CLUSTERS
REPRESENTED**3**TIER-A
RECORDS**0**EFFECT ESTIMATES
EXTRACTED

Signal of the day

A hardware day. Three of twelve records are new instruments, and one of them is the first paper in 158 records to build a device whose stated purpose is to put an aerosol and a biological endpoint in the same apparatus.

The standing gap this watch has logged since 26 July — restated in the 2–8 August weekly over 146 records — is that **not one paper coupled an instrument to a health endpoint**. Shirkhani et al. (*J Aerosol Sci*) describe MARTHA, a high-throughput multipattern aerosol deposition device built explicitly “for respiratory toxicity and health-risk assessment”; the author list (Amanatidis, Eiguren-Fernandez, Spielman) is an aerosol-instrumentation group, not a toxicology group. The record carries **no abstract in Crossref, Europe PMC or Semantic Scholar**, so it is listed and not summarised, and the claim above rests on a title and an author list. That is exactly the right amount of weight to put on it — but it is the first time the empty cell in the f4 cross-tabulation has had a candidate at all. Alongside it, Li et al. (*Photoacoustics*) close the split-instrument problem the other way: two integrating spheres serving simultaneously as scattering cavities and as a low-frequency differential Helmholtz resonator, with the connecting tube carrying a high-frequency longitudinal mode, so absorption and scattering at **450 and 532 nm** are measured *in the same volume on one sampling path* (LOD **1.2** and **1.32 Mm⁻¹** at 1 s, 3 σ). Laffolley et al. (*ACS Omega*) go the opposite direction on cost: a stereolithography-printed two-stage virtual impactor, disposable by incineration, CFD-optimised to **9.0** and **1.3 μm** cutoffs. Three different answers to the same question — what does a size- or wavelength-resolved measurement have to cost?

What changed today

- ▶ **Zero effect estimates.** The forest plot is absent from this issue for the first time since the watch began. This is a reporting-format fact, not an absence of quantitative content. Three records carry intervals — Urso et al.'s **41 % (24.4–57.4)** of EU-27 dementia DALYs attributable to modifiable risk factors, and their **7.76 M (6.69–8.72)** and **12.28 M (10.44–13.90)** prevalence counts — but a population-attributable fraction and a case count are not ratios and have no defensible position on a log axis whose null is 1.0. The same rule excluded mmHg mean differences on 8 August and an OR spanning a factor of fifteen on 7 August. Ni et al.'s $\rho = 0.28$ and AUC 0.607 are likewise not ratio-scale. Rather than render one figure dishonestly, `f5_forest` is not generated and the numbers are carried in the digest text and the register.
- ▶ **Sunday, and the thinnest window on record after yesterday's: 12 records against 14 on Saturday and 26 on Friday.** The PubMed connector returned **6 on the health axis and 4 on the instrumentation axis**, ending three consecutive empty sensing queries from that leg; a third broad MeSH sweep returned 15 for the whole day, of which 4 were already shipped in the 4 August issue and re-surfaced on re-indexing, and 4 were off-topic. The local harvester returned 5 and 1. **Crossref by ISSN across 17 journals returned 9, of which 3 carried** — a materially lower yield than the 6-of-7 it delivered on Saturday.
- ▶ **The instrumentation axis carried the issue, and two thirds of it came from outside PubMed's biomedical index.** Eight of twelve records are measurement-side (3 sensing, 3 exposure assessment, 2 occupational). The two strongest instrument papers reached the watch by different routes — Li et al. through PubMed only because *Photoacoustics* is biomedically indexed, Laffolley et al. only through Europe PMC's `CREATION_DATE` leg. Neither would have been found by the health query.
- ▶ **Three of twelve records still carry no abstract anywhere, and all three are on the measurement side.** *Atmospheric Environment* (Ray, size-resolved hygroscopicity and CCN activity), *Journal of Aerosol Science* (Shirkhani, MARTHA) and *Journal of Hazardous Materials* (Wei, size-resolved oxidative potential) — all same-day Elsevier deposits, all returning HTTP 404 from Semantic Scholar. Separately, **the six Elsevier DOIs flagged for re-check on 8 August still have no abstract in Crossref or Europe PMC 24 hours later**, so the deposit lag is longer than a day and the re-check should move to a weekly sweep rather than a daily one. Three records held from 7–8 August *did* resolve overnight (`reveh-2026-0032`, `envpo1.2026.128933`, `acsomega.6c00972`) — but all three had already shipped, so the abstracts arrived too late to be used.
- ▶ **Nothing new on calibration or drift for a second consecutive day, and the reason is now measured rather than suspected.** Three Consensus sweeps (calibration and drift, satellite–surface fusion, humidity and hygroscopic-growth correction) returned 20 hits each, capped at 3. All nine returned DOIs resolved to Crossref creation dates between **12 January and 15 July 2026** — none within four weeks of the window, let alone inside it. Consensus is a relevance index, not a recency index; it cannot see a single day and should stop being asked to. **OpenAlex returned HTTP 429 for the seventh consecutive run.** arXiv returned 50 records, none published on or after 1 August.
- ▶ **Trial watch: zero registrations or updates.** Both ClinicalTrials.gov queries (air-pollution/PM conditions; purifier, filtration, HEPA, cookstove and mask interventions) returned 0 for 9 August. `state/trials.json` is unchanged at the 21 registrations reported in the 2–8 August weekly.

Corpus analytics

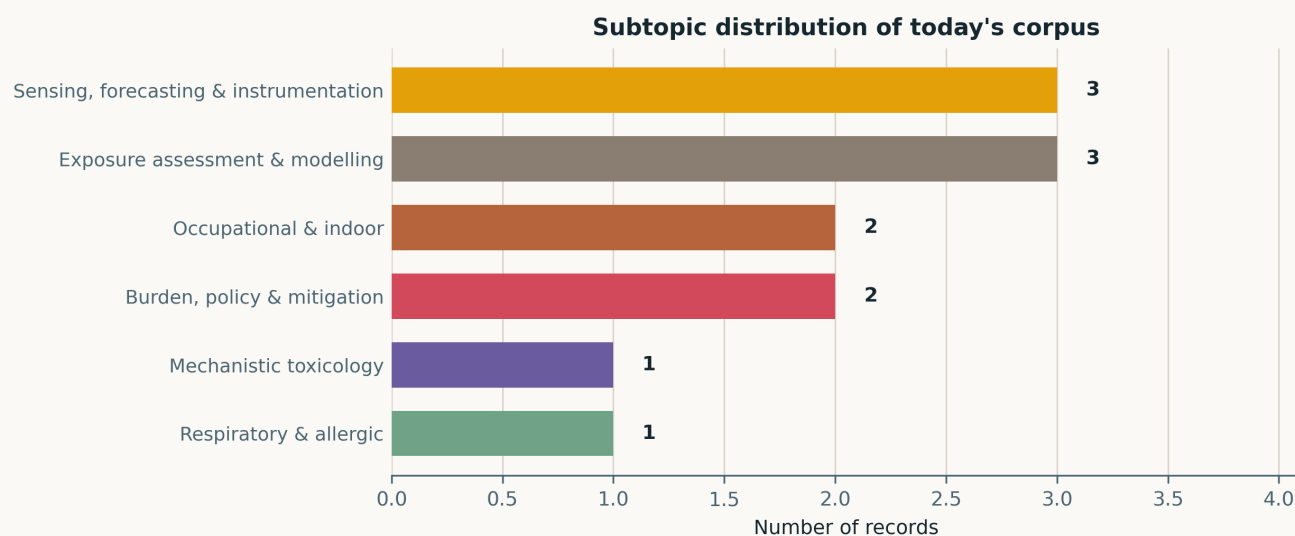


Fig. Subtopic assignment of the 12 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* and *Exposure assessment & modelling* tie at 3 and *Occupational & indoor* and *Burden, policy & mitigation* at 2; **8 of 12 records are on the measurement side, the highest share this watch has recorded**. Four of the ten canonical clusters are empty — *Neuro / mental health*, *Cardiovascular & metabolic*, *Reproductive & developmental* and *Other clinical endpoints*. The dementia-burden record is deliberately filed under *Burden* rather than *Neuro*, because its contribution is an attributable-fraction estimate, not a neurological finding.

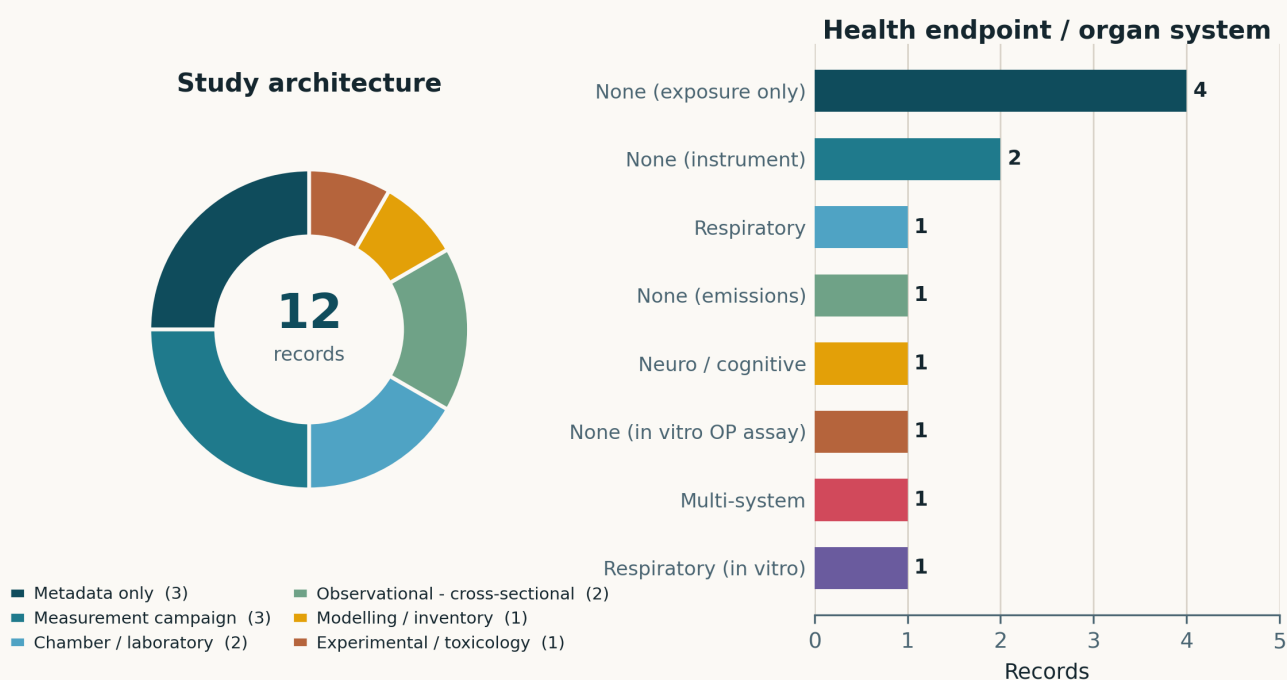


Fig. Left — study architecture. *Metadata only* and *Measurement campaign* tie at 3 of 12, with *Chamber / laboratory* at 2 and *Observational — cross-sectional* at 2; *Modelling / inventory* and *Experimental / toxicology* carry 1 each. Six architectures across twelve records is the widest spread the watch has rendered at this *n*. **No observational cohort and no controlled trial entered today**, for the second consecutive issue. Right — health endpoint. **Eight of twelve records carry no human health endpoint**: two instrument-development papers, four pure exposure-monitoring records, one *in vitro* oxidative-potential assay and one emissions bench study. The four that do are the respiratory *in vitro* deposition device, the multi-system mining survey, the neuro/cognitive burden estimate and the respiratory clinical series — so the endpoint panel describes four records and is not a field-level distribution.

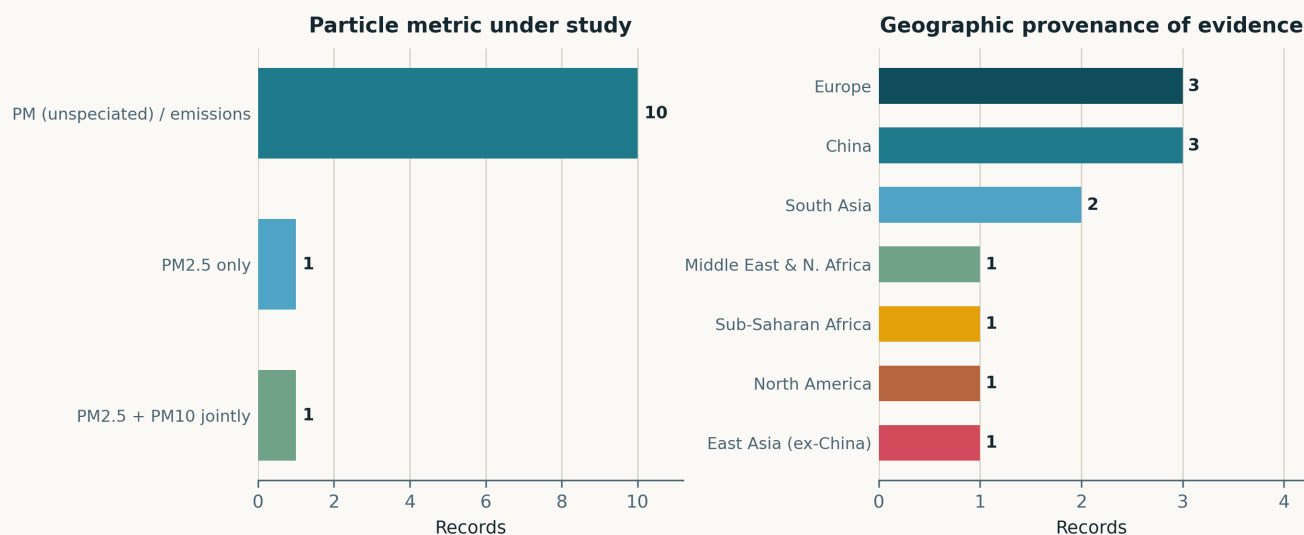


Fig. Left — particle metric, and the panel is **measuring this watch's taxonomy rather than the literature**. Ten of twelve records land in *PM (unspeciated) / emissions* against one *PM_{2.5} only* (Ni et al.) and one *PM_{2.5} + PM₁₀ jointly* (Adeniji et al.). The classifier has no bin for the quantities the day's instruments actually report — absorption and scattering coefficient, aerodynamic-diameter class, size-resolved hygroscopicity (κ), size-resolved oxidative potential, gravimetric mass with SEM morphology — so six physically distinct metrics collapse into one bar. **The figure is correct and the vocabulary is not**; this is logged as a defect against `plots.py`, not read as a finding. Right — geographic provenance. Europe and China tie at 3, South Asia 2, and Sub-Saharan Africa appears for the first time in **six issues** (Abeokuta, Nigeria; zero in each issue from 4 to 8 August). North America contributes 1, its lowest share of the watch.

Life-course windows addressed today (records may span several stages)

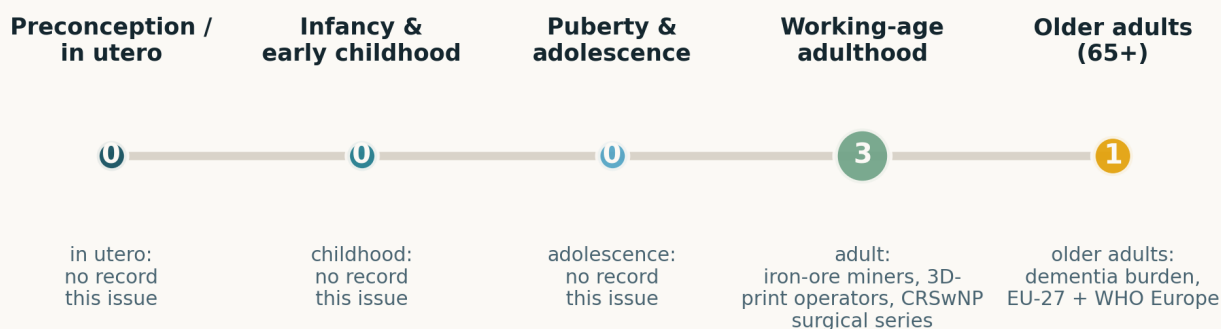


Fig. Life-course windows addressed. Counts are non-exclusive and total 4, not 12, because eight records address no human life stage at all. **The three perinatal and paediatric stages are empty for the first time in this watch**; at $n = 12$ on a Sunday this is sampling, not a field-level absence, and the reproductive/developmental literature was present as recently as 8 August.

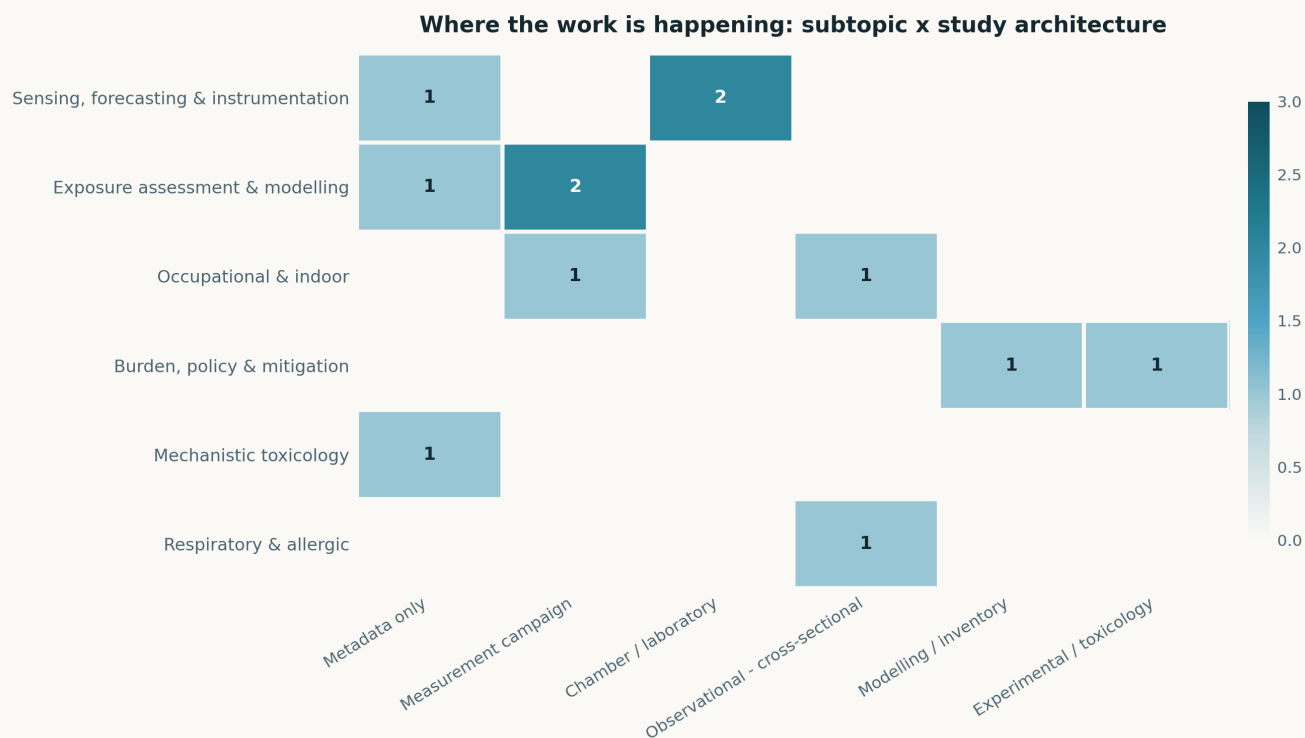


Fig. Subtopic × study-architecture cross-tabulation. Two features are worth naming. First, the *Metadata only* column is again distributed across *Sensing*, *Exposure assessment* and *Mechanistic toxicology* rather than across the health rows — the fifth consecutive issue in which the least characterisable records are the most methodologically relevant ones. Second, the *Sensing* row is populated only in *Metadata only* (1) and *Chamber / laboratory* (2) — every sensing record today is either an uncharacterisable deposit or a bench instrument, and **not one is a field co-location or a deployment**, which is the architecture a distributed network is actually built from. The instrument × health-endpoint coupling the 2–8 August weekly identified as structurally empty is still not visible here, because MARTHA is filed under *Metadata only* and cannot be scored on architecture.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked METADATA ONLY were surfaced without an abstract in Crossref, Europe PMC or Semantic Scholar and are listed, not summarised.

Sensing, forecasting & instrumentation

3 records

Li et al. 2026 Photoacoustics

Spatiotemporally synchronised dual-wavelength photoacoustic-scattering measurement of aerosol optical properties

A sphere-tube coupled resonator that removes the transport, response-time and microenvironment mismatch inherent to split absorption/scattering instruments. Two integrating spheres act simultaneously as scattered-light collection cavities and as acoustic buffers forming a low-frequency **differential Helmholtz** resonator, while the central connecting tube supports a high-frequency **second-order longitudinal** mode; 450 and 532 nm lasers are modulated at **2572** and **169 Hz** and demodulated by an FPGA digital lock-in, so both wavelengths and both quantities are recovered from one sampling path. Absorption channels calibrated against NO₂, scattering against N₂/CO₂ Rayleigh scattering; at 1 s integration and 3 σ the 532 nm limits of detection are **1.2 Mm⁻¹** (absorption) and **1.32 Mm⁻¹** (scattering). The validation is the weak part: water mist, controlled combustion and a field check establish channel isolation and feasibility but not accuracy against a reference absorption method, and no Allan-variance drift figure is given, so the 1 s LOD does not tell you the useful averaging time. For distributed sensing the transferable idea is architectural rather than economic — this is a laboratory instrument, but co-locating absorption and scattering in a shared volume is what would let a node report single-scattering albedo instead of a mass proxy.

doi: 10.1016/j.pacs.2026.100864

Laffolley et al. 2026 ACS Omega

A 3D-printed two-stage virtual impactor (μ SPLIT) for radioactive aerosol size classification and direct analysis

Stereolithography-printed virtual impactor separating aerosol into **>10, 1–10 and <1 μ m** aerodynamic classes onto integrated filters for direct post-collection α counting, designed for Fukushima Daiichi decommissioning where the instrument must be cheap enough to incinerate rather than decontaminate. Flow path optimised by CFD with Lagrangian particle tracking; predicted cutoffs **9.0** and **1.3 μ m**. The paper is unusually candid about its own failure modes: simulations identify small-particle contamination in the second-stage minor flow and large-particle contamination in the major flow, and dimensional metrology on the printed parts found channel contraction and corner rounding — i.e. the as-built geometry is not the as-designed geometry, which for an impactor moves the cutoff directly. Experimental validation is a proof of concept only: incense smoke, and Rn-progeny-bearing NaCl at 1.70 μ m GMAD putting peak α activity on the middle filter. No standardised-aerosol calibration and no quantitative wall-loss budget, both of which the authors name as required next steps. Relevant well beyond radiological work — additive manufacture of size-selective inlets is the cheapest available route to giving a low-cost node a real aerodynamic size cut instead of an optical inference.

doi: 10.1021/acsomega.6c04521

Shirkhani et al. 2026 J Aerosol Sci

Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue (S2 returns 404 on the DOI). Listed rather than summarised. It is flagged as the day's highest-priority re-check for a specific reason: the title states an aerosol-delivery instrument built for respiratory toxicity assessment, which is the exact instrument $\times$ health-endpoint coupling that the 2–8 August weekly identified as structurally absent across 146 records. Nothing beyond the title and the author list (Shirkhani, Salana, Amanatidis, Washeem, Eiguren-Fernandez, Spielman, Nannu Shankar, Lewis) supports that reading yet.

doi: 10.1016/j.jaerosci.2026.106882

Exposure assessment & modelling

3 records

Adeniji et al. 2026 PLOS Glob Public Health

Long-term air quality and meteorological coupling dynamics in Abeokuta, Nigeria

Twelve years of hourly PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ (2012–2023) against six meteorological variables at a single West African site. Particulate matter dominates and **consistently exceeds WHO guideline values**, peaking in the December–March Harmattan; regression against temperature gives R² **0.48–0.52**, with negative associations for rainfall and humidity consistent with thermal stagnation and wet scavenging. The aggregate AQI stays in the “moderate” band (39–51) while the health-risk-adjusted AQI exceeds **90** seasonally — the paper's real contribution, since it demonstrates that an index which averages across pollutants conceals a PM-driven risk that a health-weighted index exposes. Two load-bearing limitations: the data provenance for a 12-year hourly record in Abeokuta is not established in the abstract, and reanalysis or satellite-derived series are common substitutes at such sites; and single-site, single-city inference cannot separate the meteorological coupling from local source seasonality (biomass burning peaks in the same months as the Harmattan). Sub-Saharan Africa remains the most under-monitored region in this watch and this is the first record from it in six issues.

doi: 10.1371/journal.pgph.0006683

Ray et al. 2026 Atmos Environ

Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Size-resolved hygroscopicity (κ) is the physical quantity underneath every relative-humidity correction applied to an optical PM sensor, and tropical montane conditions are the high-RH regime where those corrections fail worst, so this is a priority re-check even though the paper's own framing is cloud-condensation nuclei rather than air quality.

doi: 10.1016/j.atmosenv.2026.122284

de la Torre et al. 2026 J Hazard Mater

A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain

Polyurethane-foam passive samplers at **23 sites, 2009–2023**, evaluating Stockholm Convention effectiveness. Hexachlorobenzene dominates at a median **41 %** of Σ POPs with a remote-enriched spatial pattern; legacy pesticides and indicator PCBs show temperature-dependent secondary volatilisation from contaminated soils by Clausius–Clapeyron analysis, and an *urban* enrichment plausibly driven by heat-island effects; PBDEs show a **35-fold** urban-to-remote gradient from late-regulated BDE-209. Early-banned compounds decline up to **–33 %/yr** while PeCB, iPCBs and BDE-209 are **stalled or rising to +20 %**. This is semi-volatile organic chemistry, not particulate mass, and enters on method rather than substance: a passive sampler is the cheapest long-record instrument in atmospheric science and this is a clean demonstration of what a 15-year, 23-site, zero-power network buys — an altitudinal profile on Tenerife separating local secondary emission from free-troposphere clearance. The tradeoff is equally clear: no time resolution, no size information, and partitioning between gas and particle phases is unresolved throughout.

doi: 10.1016/j.jhazmat.2026.143155

Occupational & indoor exposure

2 records

Salcines et al. 2026 J Environ Health Sci Eng

Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments

Tiered exposure assessment to EN 17058:2018 and OECD guidance around fused-deposition printing of graphene-oxide-loaded PLA, combining **near-field and far-field particle counters** with SEM and Raman speciation plus VOC measurement against Spanish and Finnish 2022 occupational limits, with ARIMA used to test whether the observed excursions are distinguishable from background drift. Nanoparticles and agglomerates were present at low average number concentration and **GO was detected in the respirable fraction**, below proposed limits but demonstrably released — the finding is qualitative (release occurred) rather than quantitative (how much). Weaknesses are structural: a pilot with no stated replicate count, number-concentration monitors that cannot speciate, so the GO attribution rests entirely on offline SEM/Raman of collected material, and occupational limits for GO that do not formally exist. The methodological point transfers directly — paired near-field/far-field counters with an offline speciation leg is the minimum viable design for attributing a number-concentration excursion to a source, and it is the design a distributed indoor network would need.

doi: 10.1007/s40201-026-00998-w

Shadmani et al. 2026 J Environ Health Sci Eng

A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce

Cross-sectional study of **1,193** Iranian iron-ore mine employees, stratified into high, moderate and low exposure *by job role*, with physical examination, clinical chemistry and haematology compared by ANOVA and χ^2 and modelled by linear and logistic regression. The workforce is young (mean **34.9 ± 7.6 y**) with low chronic-disease prevalence; the burden is localised — hearing **23.8 %**, vision **17.8 %**, dermatological **31.3 %** — while most haematological and metabolic markers did not differ by exposure group. Platelet count, LDL and blood pressure were *lower* in the high-exposure group, which the authors do not resolve and which has an obvious competing explanation the paper does not exclude: the healthy-worker survivor effect plus role-based confounding by age and seniority. The central methodological weakness for this watch is that **exposure is a job-role proxy with no measured dust concentration at all** — no gravimetric respirable sampling, no personal monitoring, no size fractionation. A three-level ordinal surrogate cannot support an exposure–response claim, and the paper’s own conclusion (localised sensory and dermatological effects, not systemic metabolic ones) is a statement about a young cohort with short cumulative exposure rather than about the hazard.

doi: 10.1007/s40201-026-01000-3

Mechanistic toxicology

1 record

Wei et al. 2026 J Hazard Mater

Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Noted because size-resolved oxidative potential is one of the few routinely measured quantities that is both a particle property and a plausible biological effect metric, and because a source-driven *seasonal* shift in OP at fixed mass concentration would be direct evidence against mass as a sufficient exposure metric — which is the substantive question behind this watch’s instrumentation axis. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.jhazmat.2026.143237

Burden, policy & mitigation

2 records

Urso et al. 2026 Lancet Reg Health Eur

Burden of dementia in Europe, 1990–2023: a systematic analysis from the Global Burden of Disease Study 2023

GBD 2023 secondary analysis for the EU-27 and the WHO European Region, with non-fatal outcomes from DisMod-MR 2.1 and dementia-attributable mortality from an excess-mortality framework. **7.76 M (95 % UI 6.69–8.72)** prevalent cases in the EU-27 and **12.28 M (10.44–13.90)** across the WHO European Region, both roughly **+90 % since 1990** despite modest declines in age-standardised prevalence; dementia moved from the eighth to the **third** leading cause of death in the EU-27. **41 % (24.4–57.4)** of EU-27 dementia DALYs are attributed to six modifiable risk factors, with **ambient particulate matter pollution named first**, ahead of high fasting plasma glucose and high BMI, and concentrated in socioeconomically disadvantaged populations. The limitation is the one every GBD risk-factor attribution carries and this paper does not resolve: the PM_{2.5}–dementia relative risk is imported from a comparative-risk-assessment function fitted to a thin and heterogeneous cohort literature, and the **24.4–57.4** interval is a modelling uncertainty, not a sampling one — it does not propagate uncertainty in the exposure surface, which for a 1990–2023 European back-cast is substantial. Treated here as a burden estimate, not as neurological evidence. The interval is deliberately absent from the forest plot: an attributable fraction has no position on a ratio axis.

doi: 10.1016/j.lanepe.2026.101796

Sathasivam & Veluchamy 2026 Environ Sci Pollut Res*Visual characteristics of PM from Halymenia dilatata microalgae biodiesel compared to low-sulfur diesel*

Single-cylinder Kirloskar CI engine, four microalgae biodiesel blends (B10–B25) against S10 low-sulfur diesel, PM collected on a textile element in a deceleration chamber, quantified gravimetrically and characterised by SEM for morphology, deposition and agglomeration. **PM mass reduction 10.44–69.79 %** across blends. The result does not vary monotonically with blend ratio — SEM showed heavier deposition for S10, B15 and B25 but lighter for B10 and B20 — which the paper reports but does not explain, and a non-monotonic dose–response across four points with no stated replication is more likely measurement noise than combustion chemistry. Three further limits matter: a textile collection element in a deceleration chamber is not a dilution tunnel and has no characterised collection efficiency as a function of size, so the gravimetric numbers are not comparable with any regulated emissions measurement; no size distribution or number concentration was measured, so a mass reduction is fully compatible with an increase in ultrafine number; and no engine load or speed conditions are given in the abstract. Included as a tier-C source-characterisation record.

doi: 10.1007/s11356-026-38134-8

Respiratory & allergic**1 record****Ni et al. 2026** World Allergy Organ J*Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps*

Multicentre retrospective series, **1,047** CRSwNP patients undergoing endoscopic sinus surgery at three tertiary hospitals in Jiangsu, with paired blood and nasal-mucosa eosinophil counts at 2008, 2012 and 2016, analysed by correlation, ROC, linear regression and linear mixed-effects models. Blood eosinophils are a poor surrogate for tissue eosinophilia ($\rho = 0.28$, **AUC 0.607**, adjusted R^2 **0.053**) and vary heterogeneously by hospital and year. City-level GDP correlated with mean blood ($\rho = 0.80$, $p = 0.014$) and tissue ($\rho = 0.72$, $p = 0.037$) eosinophil counts, whereas **annual mean PM_{2.5} was associated with neither**. The null is the reason this record is carried, and it should be read as a measurement failure rather than as evidence of no effect: the exposure is a *city-level annual mean* assigned to surgical patients at three time points, which is an ecological contrast with essentially no within-city variance, over an interval (2008–2016) in which Chinese PM_{2.5} fell steeply and uniformly — so exposure and calendar time are near-collinear by construction, and any true association is absorbed by the year term. With eight city-level observations, the GDP correlations are also fragile. Tier C.

doi: 10.1016/j.waojou.2026.101440

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Ni et al.	Socioeconomic and regional influences on eosinophilic inflammation in chronic rhinosinusitis with nasal polyps	<i>World Allergy Organ J</i>	Retrospective clinical series	Annual mean PM _{2.5} (city-level)	China
Mechanistic toxicology					
Wei et al.	Seasonal source shifts drive oxidative potential of size-resolved particulate matter in coastal southeastern China	<i>J Hazard Mater</i>	Metadata only (no abstract)	Size-resolved PM oxidative potential	China
Occupational & indoor					
Salcines et al.	Monitoring and risk assessment of airborne graphene oxide released during 3D printing with polylactic acid filaments	<i>J Environ Health Sci Eng</i>	Field measurement	Nanoparticle number conc. + respirable fraction	Spain
Shadmani et al.	A comprehensive health profile and exposure-risk analysis in an iron ore mining workforce	<i>J Environ Health Sci Eng</i>	Cross-sectional survey	Job-role dust exposure tertile (proxy)	Iran
Sensing, forecasting & instrumentation					
Laffolley et al.	A 3D-printed two-stage virtual impactor (uSPLIT) for radioactive aerosol size classification and direct analysis	<i>ACS Omega</i>	Instrument development	Aerodynamic diameter, 3 classes (>10, 1-10, <1 µm)	Japan
Li et al.	Spatiotemporally synchronized dual-wavelength photoacoustic-scattering measurement of aerosol optical properties	<i>Photoacoustics</i>	Laboratory metrology	Absorption & scattering coefficients, 450/532 nm	China
Shirkhani et al.	Design and development of a high-throughput multipattern aerosol deposition device (MARTHA) for respiratory toxicity and health-risk assessment	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Aerosol deposition (unspecified)	USA
Exposure assessment & modelling					
Adeniji et al.	Long-term air quality and meteorological coupling dynamics in Abeokuta, Nigeria	<i>PLOS Glob Public Health</i>	Monitoring-record analysis	PM _{2.5} , PM _{1.0} + 4 gases, hourly	Nigeria
Ray et al.	Evaluating CCN activity through size-resolved hygroscopicity: synergistic observations and simulations in a tropical montane region	<i>Atmos Environ</i>	Metadata only (no abstract)	Size-resolved hygroscopicity (kappa), CCN	India
de la Torre et al.	A 15-year atmospheric record of intentionally produced persistent organic pollutants in Spain	<i>J Hazard Mater</i>	Environmental surveillance campaign	Semi-volatile POPs (passive PUF samplers)	Spain
Burden, policy & mitigation					
Sathasivam & Veluchamy	Visual characteristics of PM from <i>Halymenia dilatata</i> microalgae biodiesel compared to low-sulfur diesel fuel	<i>Environ Sci Pollut Res</i>	Chamber emission characterisation	Gravimetric PM mass + SEM morphology	India
Urso et al.	Burden of dementia in Europe, 1990-2023: a systematic analysis from the Global Burden of Disease Study 2023	<i>Lancet Reg Health Eur</i>	Burden estimation (GBD)	Ambient PM (GBD risk-factor exposure)	EU-27

Provenance and method

Entry window 2026-08-09 to 2026-08-09, contiguous with the 8 August issue; no gap. Legs run: **PubMed** connector, two independent queries against [Date - Entry] — health axis 6 hits, instrumentation axis 4 (overlapping), plus a third broad sweep returning 15 for the day. **Four of those 15 were already shipped in the 4 August issue** and re-surfaced on re-indexing; they were caught by `seen.json` on both PMID and DOI. The local harvester returned 5 health and 1 sensing. **Europe PMC** on `CREATION_DATE` returned 6, of which 3 were already shipped and 1 was trade news; **2 carried**, including the day's second instrument paper. **Crossref by ISSN** across 17 journals returned 9, **3 carried**. **Consensus** ×3 sweeps returned 20 each, capped at 3 per sweep by the free tier; all 9 returned DOIs resolved to Crossref creation dates between 12 January and 15 July 2026 and were rejected as outside the window, 0 carried. **arXiv** 50 records, none published on or after 1 August. **OpenAlex** HTTP 429, seventh consecutive run. **ClinicalTrials.gov** 0 registrations or updates on both the condition and the intervention query. **16 records rejected**, each with a reason in `state/rejected.jsonl`. Three carried records have no abstract in any of the three checked sources and are listed without a summary; they are marked `METADATA ONLY` in the digest and in the architecture donut. **No forest plot is generated for this issue**: the three interval estimates in the corpus are a population-attributable fraction and two prevalence counts, none of which is ratio-scale, and `f5_forest.png` is therefore intentionally absent rather than missing. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.